

Weekly Internship Meeting

Crowd-Aware Front-Following for Assistive Mobile Robots

Marina Fragkouli
marina.frag3@gmail.com

https://github.com/marina-frag/crowd_aware_ws

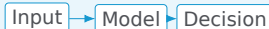
Supervisor: Antreas Vatistas

Week 1 Update July 21, 2026

Problem Description

- ▶ Front-following keeps the robot ahead of its intended user while guiding or accompanying them in the user's intended direction.
- ▶ Crowds introduce target confusion, occlusion, rapidly changing free space, and socially sensitive interactions.
- ▶ The robot must preserve formation without blocking, surprising, or endangering nearby pedestrians.

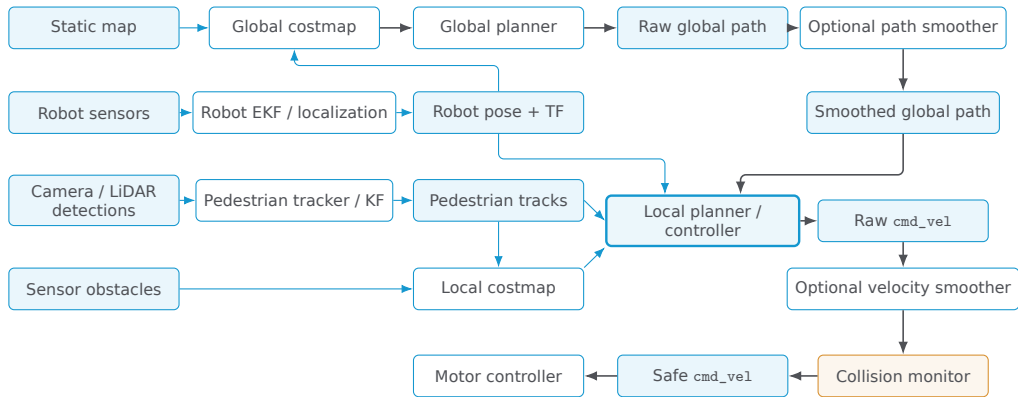
Assistive front-following in a crowd



Contribution: A Nav2-compatible, crowd-aware local planner/controller that uses semantic context to follow social protocols while keeping an assistive robot safely in front of its intended user.

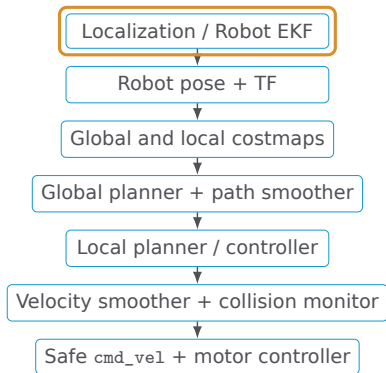
Scope: local motion generation and control; perception and tracking are upstream inputs.

Robot Navigation Pipeline 1/7



The local controller combines the global route, rolling local geometry, robot state, and identity-preserving pedestrian tracks.

Robot Navigation Pipeline 2/7

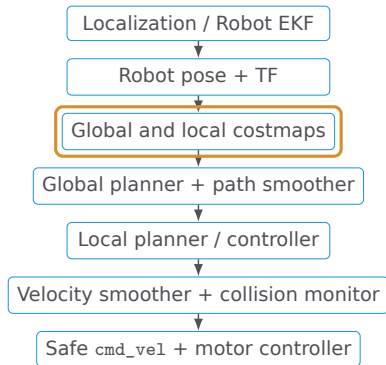


Localization

- ▶ Continuously estimate the robot pose.
- ▶ Localization publishes `map` → `odom`.
- ▶ Odometry/EKF publishes `odom` → `base_link`.
- ▶ A charging station may provide a known initial or reference pose.

Takeaway: Every planning input must use a consistent frame and timestamp.

Robot Navigation Pipeline 3/7

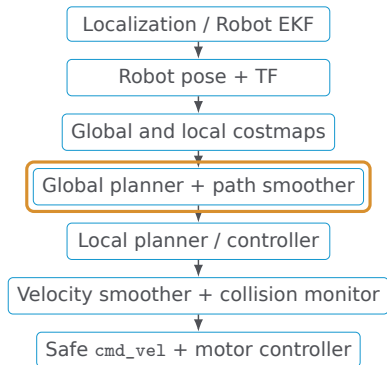


Costmaps

- ▶ **Global costmap:** large, mostly static representation for global planning.
- ▶ **Local costmap:** rolling, frequently updated representation for local control.
- ▶ Add social costs locally, but retain pedestrian IDs, velocities, predictions, and confidence as separate structured data.

Takeaway: A costmap alone cannot preserve agent identity or motion state.

Robot Navigation Pipeline 4/7

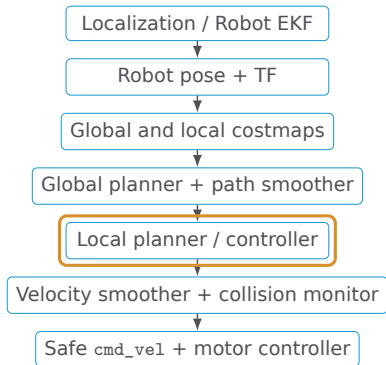


Global planner

- ▶ Compute a global path to the current desired front-following pose.
- ▶ Usually use graph search, commonly A* or a related method.
- ▶ Treat this as an existing module, not the main project contribution.
- ▶ Replan when the desired front pose moves sufficiently or the route becomes invalid.

Takeaway: The global layer chooses the route; the local layer decides how to move through people.

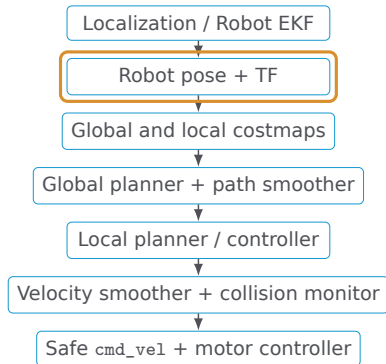
Robot Navigation Pipeline 5/7



Local planner / controller

- ▶ **Main project focus.**
- ▶ Consume the global path, local costmap, robot state, intended-user state, and individual pedestrian tracks.
- ▶ Generate or optimize feasible short-horizon trajectories.
- ▶ Combine formation keeping, crowd awareness, feasibility, progress, and smoothness.
- ▶ Produce `cmd_vel`.

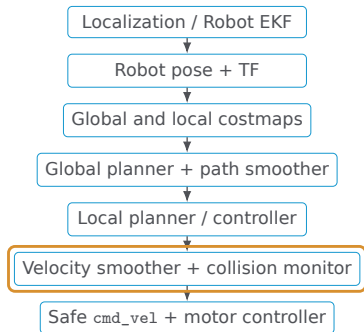
Robot Navigation Pipeline 6/7



TF / Frames

- ▶ Verify: map $\rightarrow$ odom $\rightarrow$ base_footprint/base_link $\rightarrow$ sensors.
- ▶ Transform detections and goals into the local costmap frame.
- ▶ Use base_link for robot-relative longitudinal and lateral calculations.
- ▶ A **frame** is a coordinate system; base_footprint is the ground-projected robot frame; the **footprint** is the 2D collision polygon.
- ▶ Reject or stop on stale, unavailable, or inconsistent transforms.

Robot Navigation Pipeline 7/7



Postprocessing and final safety

- ▶ **Smoother Server:** optionally modifies global-path geometry.
- ▶ **Velocity Smoother:** optionally regularizes command dynamics.
- ▶ **Collision Monitor:** enforces a final independent safety layer.

$$J_{\text{control}} = \sum_{k=1}^N [\lambda_v (\mathbf{v}_k - \mathbf{v}_{k-1})^2 + \lambda_\omega (\omega_k - \omega_{k-1})^2]$$

Path smoothing, velocity smoothing, and collision monitoring act at different stages and solve different problems.

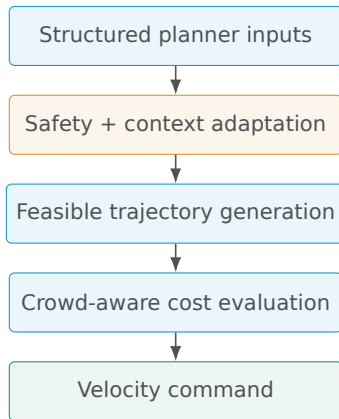
Our Extension

- ▶ Extend the Nav2 local planner/controller rather than replacing the complete stack.
- ▶ Add crowd-aware and formation-aware scoring while reusing localization, maps, global planning, command postprocessing, and collision monitoring.

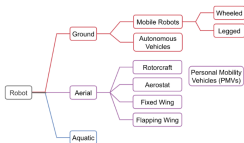
$$\tau^* = \arg \min_{\tau \in \mathcal{T}_{\text{valid}}} J(\tau)$$

Inputs: user state, desired front pose, bystander tracks, robot state, local costmap, and global path.

Output: safe, smooth, socially aware cmd_vel.

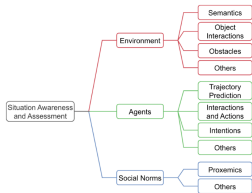


System Taxonomies & Evaluation Framework



Taxonomy based on robot type

Leaves mostly mutually exclusive

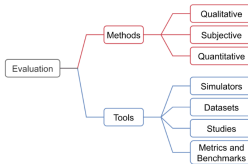


Taxonomy based on situation awareness and assessment



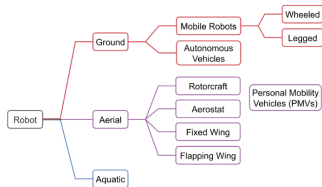
Taxonomy based on planning and decision-making

The facets are not mutually exclusive



Taxonomy for tools and evaluation methods

Taxonomy Based on Robot Type

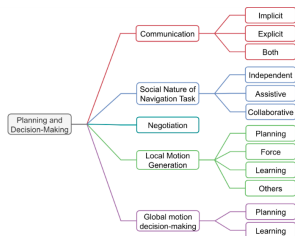


- ▶ 2D assistive mobile ground robot on approximately planar ground.
- ▶ 2D costmaps and a 2D collision footprint.
- ▶ Planar state and velocity control:

$$\mathbf{x} = [x, y, \theta]^T, \quad \mathbf{u} = [v, \omega]^T$$

Excludes aerial, underwater, legged, and full 3D motion planning.

Taxonomy Based on Planning and Decision-Making 1/6



Communication

Implicit motion first; explicit signaling later.

Communication options

- **Implicit:** trajectory, speed, posture, gaze.
- **Explicit:** speech, gestures, displays, lights.
- **Hybrid:** predictable motion supported by a clear signal.

Social nature of the task

Task type	Decision
Independent navigation among people	Baseline / background
Assistive navigation	Core
Collaborative navigation	Limited core behavior
Minimal collaboration occurs when the robot yields, waits, or adapts formation so people can pass safely.	

Taxonomy Based on Planning and Decision-Making 2/6

Negotiation

- ▶ **In scope:** implicit, rule-based yielding, slowdown, path adjustment, waiting, and stopping.
- ▶ **Out of scope:** speech-based or complex learned negotiation.

The robot communicates priority through legible motion.

Behavior priority

1. Avoid collisions; satisfy hard safety constraints.
2. Stop when user identity or state is unreliable.
3. Respect flow, personal space, and groups.
4. Preserve front-following formation.
5. Keep motion smooth and predictable.
6. Minimize delay and path length.

The order informs both hard constraints and soft-cost tuning.

Taxonomy Based on Planning and Decision-Making 3/6

Local motion generation is the project focus.

Method	Core idea	Relevance
DWA	Sample dynamically reachable velocity commands over a short horizon.	Conceptual baseline.
DWB	Generate trajectories and score them with configurable critic plugins.	Nav2-native critic extension path.
TEB	Jointly optimize timed poses for time, obstacles, and kinodynamic constraints.	Optimization foundation.
HATEB	Add time-dependent human-aware constraints and predicted human motion.	Preferred crowd-aware direction.

$$\mathcal{B} = \{p_0, \Delta t_0, p_1, \Delta t_1, \dots, p_N\}$$

TEB poses are optimization points, not mandatory stops.

Taxonomy Based on Planning and Decision-Making 4/6

DWB: critic-based sampling

$$J_{\text{DWB}}(\tau) = \sum_{i=1}^{N_c} s_i C_i(\tau)$$

$$\tau^* = \arg \min_{\tau \in \mathcal{T}_{\text{valid}}} J_{\text{DWB}}(\tau)$$

C_i : critic score; s_i : configured scale.

- ▶ No single fixed objective: enabled critics are configurable.
- ▶ Typical terms: obstacle, path alignment/distance, goal alignment/distance, rotate-to-goal, oscillation, and forward preference.

Decision: Add social and formation critics while retaining the DWB architecture as a baseline.

HATEB: graph optimization

$$J_{\text{HATEB}}(\mathbf{z}) = \sum_k \mathbf{e}_k(\mathbf{z})^T \Omega_k \mathbf{e}_k(\mathbf{z})$$

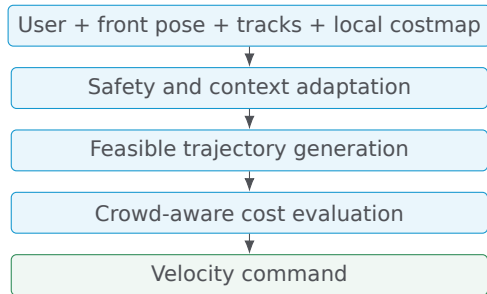
- ▶ $\mathbf{z}$: optimized poses and time differences.
- ▶ $\mathbf{e}_k$: constraint residual.
- ▶ Ω_k : information matrix / weight.

Representative terms

- ▶ velocity, acceleration, kinematics;
- ▶ execution time and path length;
- ▶ static and dynamic obstacles;
- ▶ via points and visibility;
- ▶ robot-human safety and relative velocity.

Decision: Extend HATEB for crowd-aware front-following, building on CoHAN/HATEB.

Taxonomy Based on Planning and Decision-Making 6/6



Initial direction: crowd-aware HATEB/CoHAN extension; DWB with social critics as the main baseline or fallback.

Hard constraints

- ▶ no collision with static obstacles or pedestrians;
- ▶ velocity and acceleration limits;
- ▶ controlled stop for stale data, low confidence, or no safe trajectory.

Out of scope

- ▶ force-based local motion;
- ▶ reinforcement learning;
- ▶ global motion generation.

Context-Dependent Use Cases

Context	Desired behavior
Open area	Maintain the nominal forward distance.
Narrow corridor	Align with the corridor and reduce lateral offset.
Doorway	Restrict overtaking and wait when passage is unsafe.
Junction	Increase emphasis on short-horizon user-intention prediction.
Dense crowd	Reduce speed and shorten formation distance within safe bounds.
Occlusion zone	Increase caution and rely on short-term target prediction.

Decision: The desired front-following pose is context dependent rather than fixed.

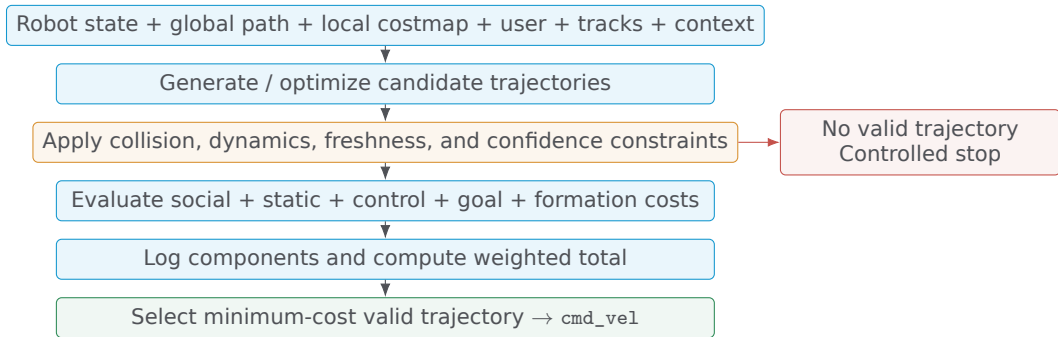
An Example of Local Motion Generation 1/2

$$J = w_s J_{\text{social}} + w_o J_{\text{static}} + w_c J_{\text{control}} + w_g J_{\text{goal}} + w_f J_{\text{formation}}$$

Cost	Role
J_{social}	Penalize intrusion into anisotropic, speed-dependent pedestrian and group space.
J_{static}	$\sum_k C_{\text{costmap}}(p_k)$; penalize walls, furniture, and mapped obstacles.
J_{control}	Penalize acceleration, braking, and rapid steering changes.
J_{goal}	Reward route progress; discourage unnecessary stalling or reversal.
$J_{\text{formation}}$	Maintain the context-dependent pose in front of the intended user.

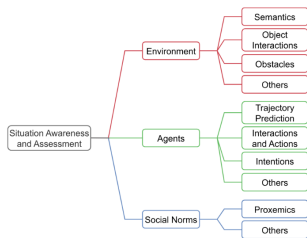
Hard safety constraints are enforced before soft-cost comparison.

An Example of Local Motion Generation 2/2



Takeaway: Choose the best valid motion; never choose the least-bad unsafe motion.

Taxonomy Based on Situation Awareness and Assessment 1/4



Keep for this project

- ▶ static obstacles and local free space;
- ▶ local crowd density;
- ▶ available passage width / constrained-space indication;
- ▶ occlusion and stale-input indication.

Do not require

- ▶ full scene semantics;
- ▶ human-object interaction recognition;
- ▶ classification of every room, door, and object.

Decision: Derive initial context from geometry and tracked agents, without a new semantic-perception model.

Taxonomy Based on Situation Awareness and Assessment 2/4

Intended user

Persistent ID
Pose and heading
Velocity
Short-term predicted state
Tracking confidence

Other pedestrians

Individual tracking ID
Pose and heading
Velocity
Short-term predicted state
Optional group label

Required representation

A collective pedestrian mask is insufficient; bystanders require individual tracks for prediction and social scoring.

Prediction choice

Constant velocity / heading, optionally Kalman-filtered. Learned RNN/LSTM/GNN prediction is out of scope.

Taxonomy Based on Situation Awareness and Assessment 3/4

Temporary target-loss policy

1. Retain the last intended-user ID and predicted state.
2. Never switch merely because another person is nearer.
3. Reduce speed as confidence falls.
4. Accept re-identification from the tracker.
5. Stop beyond confidence, prediction-age, or risk thresholds.

Interactions and groups

- ▶ Do not plan through a detected group or cut an interaction.
- ▶ Do not treat an intervening pedestrian as the new user.
- ▶ Avoid separating the user from a known companion.

First version: supplied group labels or a lightweight proximity-and-similar-velocity rule; learned group recognition is out.

Taxonomy Based on Situation Awareness and Assessment 4/4

Short-horizon intention cues

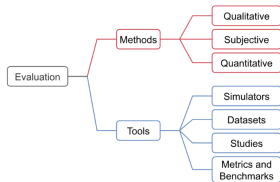
- ▶ continued straight motion;
- ▶ left or right turn;
- ▶ slowing or stopping;
- ▶ likely path crossing;
- ▶ temporary target loss.

Infer approaching and obstacle deviation from predicted trajectories, not separate learned classes.

Proxemic model	Decision
Circular fixed radius Anisotropic, heading dependent Speed-dependent size Larger joint group field	Baseline Core Core Optional / core with labels

Social cost is larger in front of a moving pedestrian, smaller behind or laterally, and grows with speed. Formation handles the user, with a hard minimum distance.

Taxonomy for Tools and Evaluation Methods 1/5



Qualitative + quantitative +
user-centered evaluation

Intended-user tracking

Metric	Measures
Target tracking success rate	Fraction of time with the correct target.
ID switch count	Incorrect changes to another pedestrian.
Target loss rate	Frequency of intended-user loss.
Reacquisition time	Time required to restore the correct track.
Track continuity	Uninterrupted duration of correct tracking.
Target-position error	Intended-user position error.
Target-heading error	Intended-user heading error.

Taxonomy for Tools and Evaluation Methods 2/5

Front-following quality

Metric	Measures
Longitudinal formation error	Error from the desired forward distance.
Lateral formation error	Sideways error from the desired formation.
Robot-user distance variance	Stability of separation over time.
User visibility rate	Fraction of time with a clear view or valid track.
Formation recovery time	Recovery time after a disturbance.
User stop count	Times the user must stop because of robot behavior.

Takeaway: These metrics distinguish “reached the goal” from “guided the user well.”

Taxonomy for Tools and Evaluation Methods 3/5

Safety and crowd awareness

Metric	Measures
Collision rate	Collisions per run or unit time.
Minimum pedestrian distance	Closest robot-pedestrian separation.
Time-to-collision	Dynamic collision risk.
Social-space intrusions	Number of personal-space violations.
Time inside social spaces	Duration of proxemic violations.
Emergency stops	Abrupt safety-layer interventions.
Near-miss count	Unsafe encounters without contact.
Group-splitting events	Robot passages through social groups.

Taxonomy for Tools and Evaluation Methods 4/5

Motion quality

Metric	Measures
Path length	Route efficiency.
Completion time	Task duration.
Linear/angular acceleration	Command smoothness.
Jerk	Abrupt changes in acceleration.
Cumulative heading change	Zigzagging and unnecessary turning.
Freeze duration/count	Robot-freezing behavior.
Replanning frequency	Planner stability.

Takeaway: Report safety and formation with efficiency; efficiency alone is insufficient.

Small user study

- ▶ perceived safety and comfort;
- ▶ trust and predictability;
- ▶ confidence that the robot knows where it is going;
- ▶ confusion and effort required to follow;
- ▶ preference between planners.

Evaluation design

- ▶ Pair subjective ratings with objective logs from the same scenario.
- ▶ Prefer within-subject planner comparisons where practical.
- ▶ Begin only after controlled simulation and technical safety validation.

Immediate Experiment Plan 1/2

Ground truth: use visible ID badges to identify the intended user.

1. One user, no pedestrians.
2. One pedestrian crossing the route.
3. Similarly dressed pedestrians near the user.
4. Temporary full occlusion.
5. Pedestrian between robot and user.
6. Narrow corridor.
7. Junction where the user turns.
8. Doorway.
9. Group blocking part of the route.
10. Rapid change in user speed or direction.

Immediate Experiment Plan 2/2

Planner comparison

1. Front-following without crowd awareness.
2. Conventional obstacle avoidance.
3. Proxemic costs.
4. Proposed confidence-, prediction-, and crowd-aware method.

Hold scenario geometry, pedestrian scripts, initial conditions, and safety limits constant.

Log and evaluate

- ▶ IDs, target state, selected trajectory;
- ▶ component costs, commands, safety events;
- ▶ target retention and formation error;
- ▶ safety, motion quality, completion;
- ▶ failure mode.

Requirements Summary 1/4

Inputs from the existing system

- ▶ robot pose, velocity, footprint, local costmap, global path;
- ▶ intended-user ID, pose, heading, velocity, prediction, confidence;
- ▶ individual bystander tracks and optional group labels;
- ▶ desired front pose or data required to update it.

Implemented in this project

- ▶ crowd-aware local trajectory generation;
- ▶ social, formation, obstacle, progress, and smoothness costs;
- ▶ context-dependent speed and formation;
- ▶ safe behavior under uncertainty or infeasibility;
- ▶ controlled non-social baseline comparison.

Requirements Summary 2/4

- FR-1 Drive toward a configurable pose in front of the intended user.
- FR-2 Continuously generate dynamically feasible, collision-free local motion.
- FR-3 Score static obstacles, pedestrian social space, formation error, progress, and smoothness.
- FR-4 Use anisotropic, heading- and speed-dependent pedestrian costs; retain a circular baseline.
- FR-5 Adapt speed and formation within configured bounds in crossings, crowds, constrained passages, and low-confidence states.

Requirements Summary 3/4

- FR-6 Avoid passing through detected groups when group information is available.
- FR-7 Retain the provided user ID; never substitute a bystander using nearest-person proximity alone.
- FR-8 Reduce speed on stale or uncertain user input; stop beyond configured confidence, risk, or feasibility thresholds.
- FR-9 Avoid unnecessary oscillation, abrupt turns, and repeated stopping.

Takeaway: Functional behavior must remain compatible with the surrounding Nav2 pipeline and independent safety layers.

Requirements Summary 4/4

Quality	Requirement
Real-time	Update the stack frequently enough for safe operation; do not act on materially outdated state.
Safety	Fail safely on unreliable perception, tracking, TF, or planning; enforce motion limits.
Modularity	Keep perception, tracking, prediction, planning, and control replaceable.
Configurability	Expose cost weights, proxemic parameters, formation settings, context thresholds, and limits.
Logging	Record inputs, trajectories, costs, commands, tracking state, and safety events.

Decision: Communicate intent first through predictable motion; explicit projection or display can later clarify ambiguous maneuvers.